

Florence: Infrastructure - Firebase (App Distribution)

1. Overview and Purpose

Firebase App Distribution serves as the primary platform for distributing pre-release Android application builds to beta testers and the client. It allows the development team to gather feedback and conduct User Acceptance Testing.

2. Configuration and Secrets Management

The continuous integration pipeline utilises specific Firebase credentials to automatically upload compiled APK files after a successful build.

Environment Variable	Description	Where it is stored	Required By
FIREBASE_APP_ID	The unique identifier for the Android application.	GitHub Secrets	GitHub Actions CI/CD
FIREBASE_SERVICE_ACCOUNT_JSON_BASE64	Base64 encoded service account key for secure API authentication.	GitHub Secrets	GitHub Actions CI/CD
FIREBASE_TOKEN	CLI token used for automated hosting or distribution tasks.	GitHub Secrets	GitHub Actions CI/CD

3. Technical Implementation Details

Project Architecture

- Project ID:** swinburne-cos40005-florence
- Project Number:** 609541142807
- Target Platform:** Android
- Package Name:** com.vanq.florence
- App ID:** 1:609541142807:android:5fd8a46d5d6a31e2b1df52

Tester Management

Testers are organised into groups to streamline the distribution process and ensure the correct builds reach the appropriate stakeholders.

- Tester Groups:** A primary group named testers is currently configured containing 6 members.
- Invite Links:** An unrestricted invite link (https://appdistribution.firebase.dev/i/3eca3fb84c514316) is active. This allows new testers to join the testing programme simply by clicking the link without requiring manual approval from the Firebase project owner.

Automated Distribution

When code is merged into the production branch and a full release is triggered, the GitHub Actions workflow compiles the Android APK and automatically uploads it to Firebase App Distribution. The release is instantly routed to the testers group and an email notification is sent to all members prompting them to update or install the application via the Firebase App Distribution tester app.

4. Billing Limits and Day 2 Operations

- Current Tier:** Free (Spark Plan). Firebase App Distribution is currently free to use for distributing pre-release apps to testers.
- Adding New Testers:** A new tester is added by sharing the unrestricted invite link or manually adding their email address to the testers group in the Firebase Console.
- Domain Restrictions:** The console currently warns that invitation restrictions cannot be enabled because the active invite link lacks domain restrictions. If the development team

wishes to restrict testing to a specific corporate email domain in the future, they must delete the current unrestricted link and generate a new one with domain limitations applied.

- **Day 2 Operations:** Testers should install the Firebase App Distribution application on their Android devices. When a new build is pushed via the CI/CD pipeline, the tester will receive a notification.